

Prompt Engineering Worksheet

RAG and Prompt Engineering: From Concepts to Creation — Module 2, Chapter 3

How to use this worksheet

For each challenge, read the weak prompt, identify what's missing, then rewrite it using the four elements:

Task · Context · Format · Constraints. There are no perfect answers — but there are definitely better ones. Try your improved prompts in any LLM and compare the outputs.

Challenge 1 Too Vague — clowns

You want to know: Whether clowns are a good choice for your kid's birthday party.

■ Weak prompt	■ What to aim for
"Tell me about clowns."	<i>Specific audience, a clear angle, a format, and a length limit.</i>

What's missing from the weak prompt?

Your improved prompt:

Hint: Who is this for? What aspect of clowns? What format? How long?

Challenge 2

Too Much At Once — goth teenager

You want to know: How to make the American Revolution interesting for your goth teenager.

■ Weak prompt	■ What to aim for
"Summarize the American Revolution, translate it to French, rewrite it for a goth teenager who hates history, add three discussion questions, and suggest a dramatic title."	Pick the ONE output you need most. Do it well. Then ask for the next thing.

Which single task is most important right now?

Your focused prompt (one task only):

Hint: Sequential prompts cost 30 seconds. Bad outputs cost a full rewrite.

Challenge 3

No Format Specified — recipe

You want to know: What to make for dinner tonight with what's in your fridge.

■ Weak prompt	■ What to aim for
"Give me a recipe."	Tell it your ingredients, how many people, how much time, and what format you want.

List your actual ingredients:

Your improved prompt:

Hint: Numbered steps? Paragraph? Shopping list? Pick one and say so.

Challenge 4 Forgetting the Audience — GLP-1

You want to know: What GLP-1 medications actually do in the body.

■ Weak prompt	■ What to aim for
"Explain how GLP-1 works."	Add exactly who is reading this. The whole answer changes.

Version A — for a newly diagnosed Type 2 patient:

Version B — for a fitness enthusiast curious about Ozempic:

Version C — for an endocrinologist reviewing patient materials:

Hint: Same question, three completely different answers needed.

Challenge 5 Not Iterating — hurling

You want to know: The rules of hurling so you can follow the match you're about to watch.

■ Weak prompt	What came back
"Tell me about hurling."	A thorough and confident guide to managing nausea and preventing vomiting.

Follow-up prompt to fix the topic:

The rules came back too long and technical. Write one follow-up to fix that:

Hint: You don't need to start over. One targeted instruction at a time.

Challenge 6

Bonus: Put It All Together — climate change

You want to write: A compelling piece about climate change for your company newsletter.

■ Weak prompt	■ Your goal
"Write something about climate change."	Build a strong prompt using all four elements: Task · Context · Format · Constraints

Task — what specifically do you want it to write?

Context — who is the audience? What's the publication? What's the angle?

Format — how long? What structure? What style?

Constraints — what tone? What to avoid? Any exclusions?

Your assembled prompt:

The Fix Framework

When a response misses, ask these three questions before re-prompting:

1	What did I not specify?	Task, audience, angle, length
2	What context did I assume the model had?	Background, role, situation
3	What format did I expect but not request?	Structure, length, style

Quick Reference: The Five Mistakes

Mistake	Symptom	Fix
Too vague	Generic, unfocused output	Add audience, angle, length
Too much at once	Everything done badly	One task at a time
No format	Wrong structure	Specify the output shape
Wrong audience	Right answer, wrong room	Say who's reading it
Not iterating	Accepting a mediocre first response	One targeted follow-up at a time